

Changliang Zhou

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Profile

I am a Ph.D. student at Southern University of Science and Technology, working on AI for Optimization, specifically Neural Combinatorial Optimization (NCO). My research focuses on deep reinforcement learning for vehicle routing problems (VRPs), with an emphasis on **out-of-distribution zero-shot generalization**. As a first author, I proposed ICAM, an instance-conditioned adaptation model for generalization to **thousand-scale** VRP instances, L2R, a neural solver scalable to **10-million-node** instances, and URS, a generalist neural solver covering **over 100** VRP variants. I also lead EasyCO, a general-purpose combinatorial optimization platform that currently standardizes workflows for about **50** solvers and supports automated solving for about **150** combinatorial optimization problems.

Education

Qingdao University , M.Sc. degree in Software Engineering, College of Computer Science and Technology	Sep. 2019–Jun. 2022
Southern University of Science and Technology , Ph.D. student in Intelligent Manufacturing and Robotics, School of Automation and Intelligent Manufacturing	Sep. 2022–Present

Highlights

ICAM: Instance-Conditioned Adaptation Model for Thousand-Scale Generalization [IEEE T-ITS 2026] Code & Paper - Introduced a lightweight instance-conditioned adaptation function and a low-complexity attention mechanism to adjust policies using instance-specific information. - Validated the method on four representative routing problems (TSP, CVRP, ATSP & CVRPTW), improving generalization while preserving fast inference.	2023–2024
L2R: Learning to Reduce Search Space for Million-Scale Generalization [KDD 2026] Code & Paper - Proposed the first learning-based framework to dynamically prune the search space by adaptively prioritizing nodes based on problem-specific features and states. - Developed the first neural solver for 10-million-node VRP instances. Trained only on 100-node instances, it generalizes to TSP10M in a zero-shot manner with a 5.05% gap.	2024–2025
URS: A Unified Neural Routing Solver for Cross-Problem Zero-Shot Generalization [ICML 2026] Code & Paper - Proposed a unified data representation that eliminates the need for problem enumeration, significantly broadening problem coverage by projecting diverse VRP variants into a unified feature space. - Developed the first neural solver that efficiently solves over 100 VRP variants with a single model without retraining or fine-tuning, while successfully scaling to 7,000-node instances.	2025–2026
EasyCO: A Learning-Driven Platform for Combinatorial Optimization [In Development] (Lead) - Established a standardized pipeline spanning data generation, environment setup, training, and evaluation. - Currently supports standardized workflows for over 50 solvers and automated solving for over 150	2025–Present

Publications

As of June 15, 2026, I have published **11 papers**, including **4 first/co-first-author papers**. My work has appeared in venues including **ICML, KDD, and IEEE T-ITS**, and has received **256 citations** on Google Scholar. Selected publications are listed below:

- **Changliang Zhou**, Canhong Yu, Shunyu Yao, Xi Lin, Zhenkun Wang[†], Yu Zhou, Qingfu Zhang. URS: A Unified Neural Routing Solver for Cross-Problem Zero-Shot Generalization. *43rd International Conference on Machine Learning (ICML)*, 2026.
- **Changliang Zhou***, Xi Lin*, Zhenkun Wang[†], Qingfu Zhang. Learning to Reduce Search Space for Generalizable Neural Routing Solver. *32nd SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2026.
- **Changliang Zhou***, Xi Lin*, Zhenkun Wang[†], Xialiang Tong, Mingxuan Yuan, Qingfu Zhang. Instance-Conditioned Adaptation for Large-Scale Generalization of Neural Routing Solver. *IEEE Transactions on Intelligent Transportation Systems (T-ITS)*, 2026.
- Zhi Zheng, **Changliang Zhou**, Xialiang Tong, Mingxuan Yuan, Zhenkun Wang[†]. UDC: A Unified Neural Divide-and-Conquer Framework for Large-Scale Combinatorial Optimization Problems. *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- Yubin Xiao, Di Wang, Boyang Li, Mingzhao Wang, Xuan Wu, **Changliang Zhou**, You Zhou[†]. Distilling Autoregressive Models to Obtain High-Performance Non-autoregressive Solvers for Vehicle Routing Problems with Faster Inference Speed. *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2024.
- Canhong Yu, **Changliang Zhou**, Rongsheng Chen, Zhenkun Wang[†], Yu Zhou. Rethinking Constraint Awareness for Efficient State Embedding of Neural Routing Solver. *arXiv preprint arXiv:2605.10122*, 2026.
- Rongsheng Chen, **Changliang Zhou**, Canhong Yu, Yuanyao Chen, Yu Zhou, Zhuo Chen, Zhenkun Wang[†]. SPACE: Unifying Symmetric and Asymmetric Routing Problems for Generalist Neural Solver. *arXiv preprint arXiv:2605.24484*, 2026.
- Yunpeng Ba, Xi Lin, **Changliang Zhou**, Ruihao Zheng, Zhenkun Wang[†], Xinyan Liang, Zhichao Lu, Jianyong Sun, Yuhua Qian, Qingfu Zhang. Survey on Neural Routing Solvers. *arXiv preprint arXiv:2602.21761*, 2026.

* Equal contribution [†] Corresponding author _____ Co-mentored student

Service and Honors

- **Reviewer**: ICLR 2026; NeurIPS 2026.
- **University Honor**: Outstanding Postgraduate Student, Southern University of Science and Technology, 2022–2023.